DeepCyan

V2ISIM

V2I Sim - Traffic Management

MEMBERS

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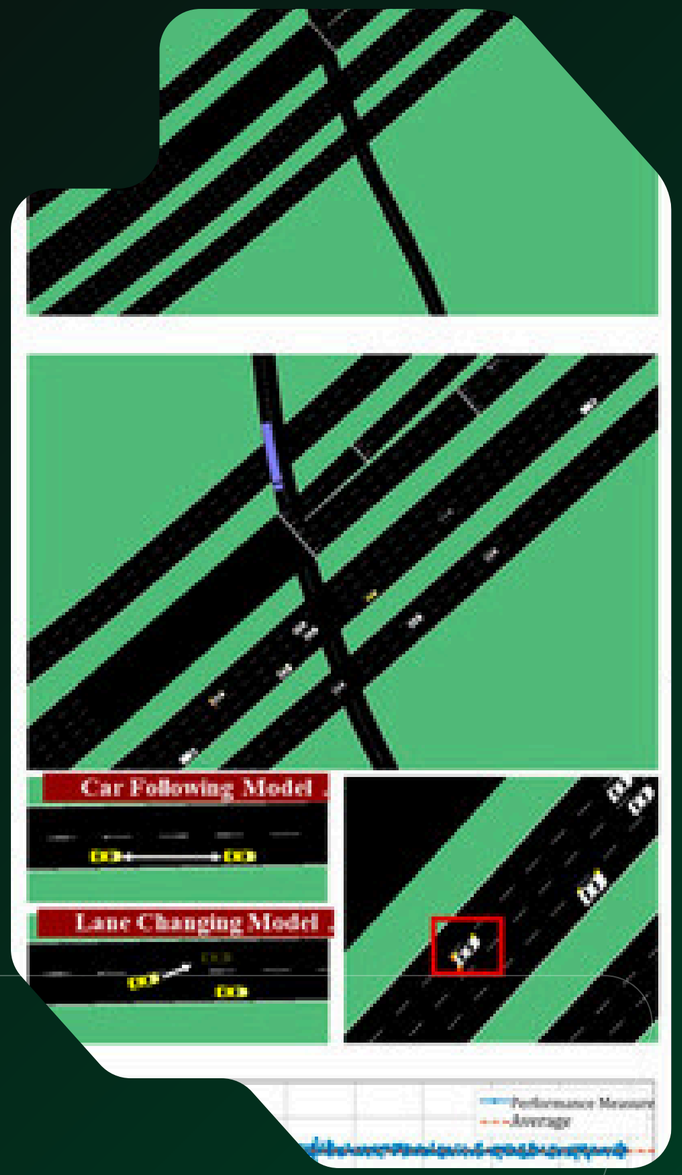
V2I Smart Traffic Management System

PROBLEMS WE ARE ADDRESSING

- Lack of coordinated vehicle–infrastructure communication
- Lack of predictive traffic management
- Testing scenario that might occur in real life
- Inability to handle dynamic road conditions
- Limited support for autonomous and connected vehicle
- Disconnected smart city systems
- Detection of anomalies



OUR SOLUTION

Our project replaces rigid traffic systems with an intelligent system that learns to optimize vehicle flow through continuous trial and error.

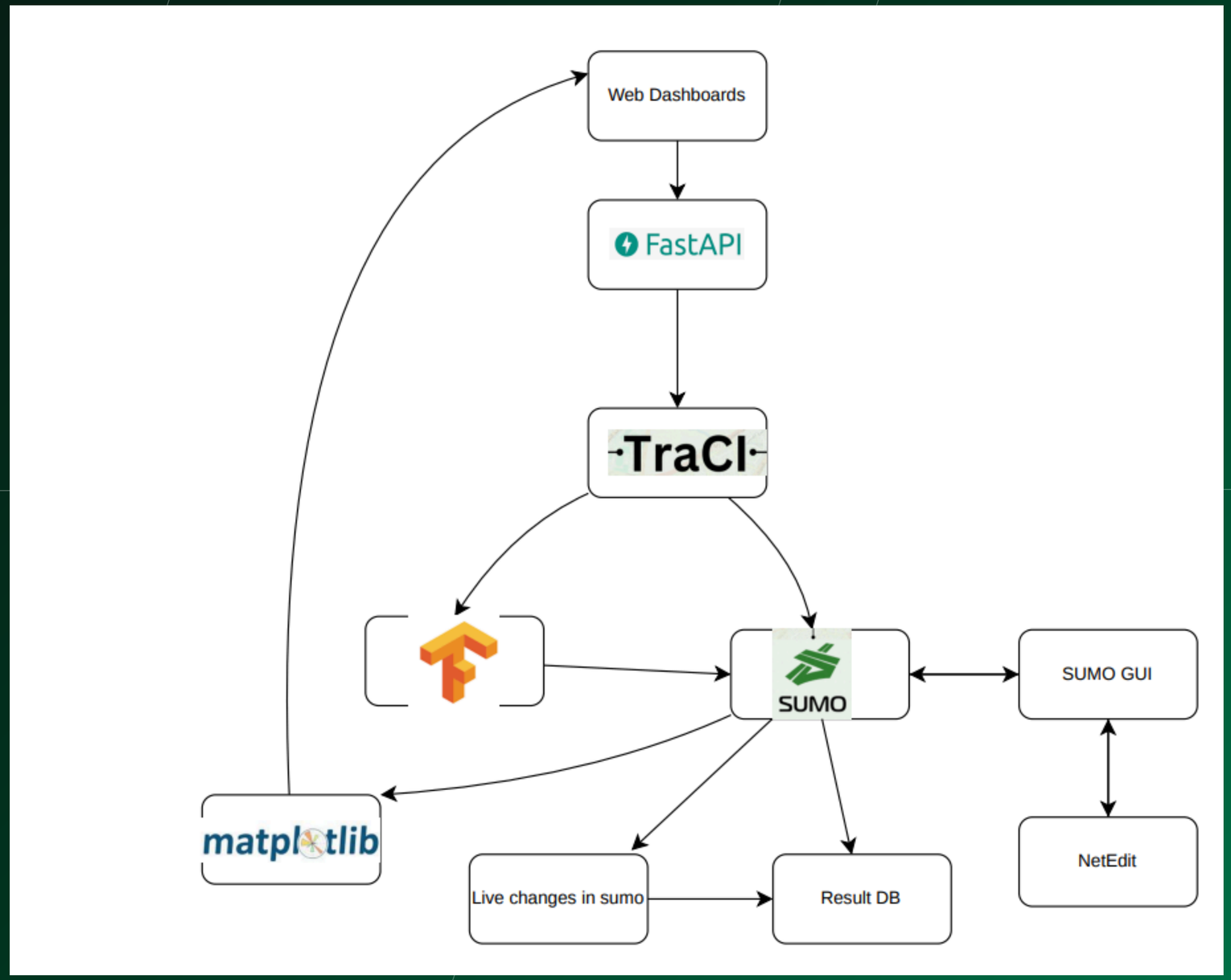
By using a simulated "Digital Twin" environment, we aim to reduce urban delays and scale our solution from single junctions to entire city blocks.

This creates a portable, data-driven framework for smarter city planning and more efficient traffic management.

KEY FEATURES

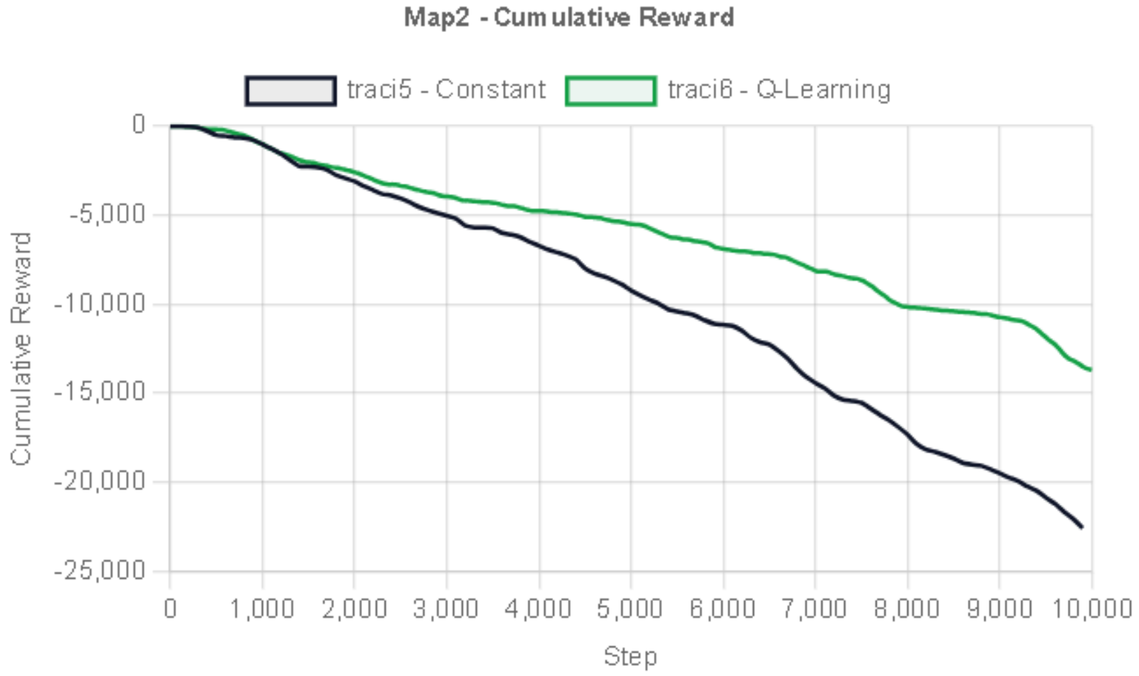
- Digital Twin of a Street
- Reinforcement Learning Control Logic
- Deep Neural Network Approximation
- Interactive Web Dashboard Control
- Automated Performance Analytics

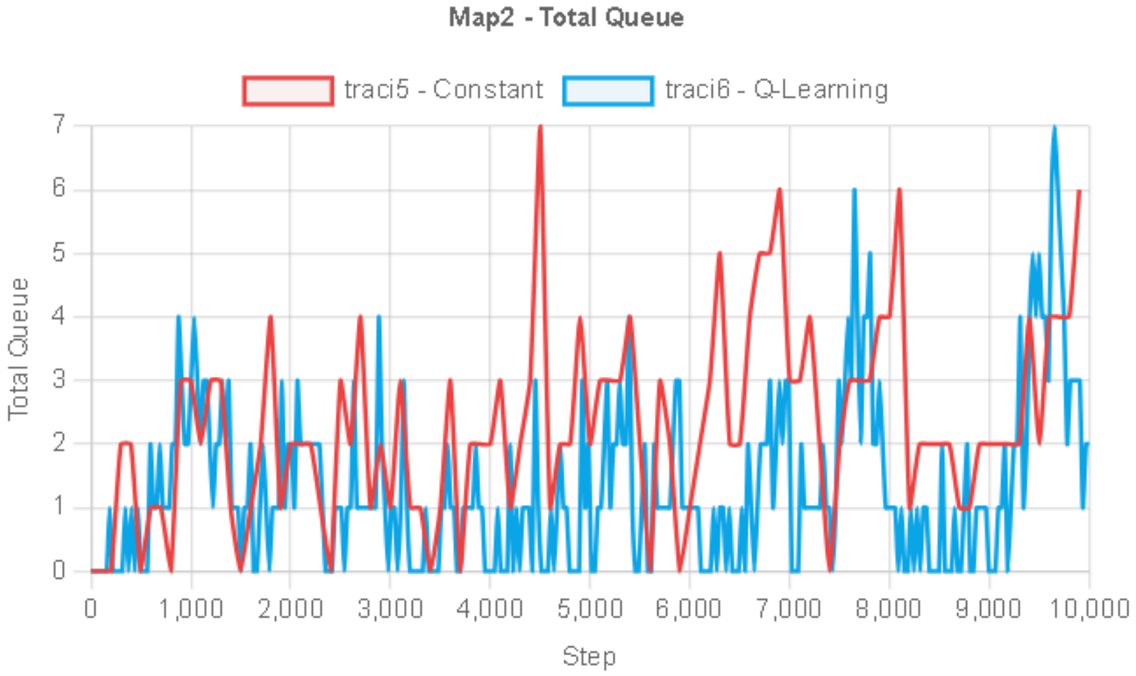
SYSTEM ARCHITECTURE DIAGRAM



TEST RESULTS

traci5 Avg Queue	traci6 Avg Queue	traci5 Final Reward	traci6 Final Reward
2.28	**1.37**	**-22577.00**	**-13692.00**





C:\WINDOWS\system32\cmd.

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Site: CSIT321 (DB25) Project | U

delias0247-lab/DeepCyan-V2lsi

DeepCyan · Graph

127.0.0.1:8000/graph

DeepCyan

- Dashboard
- Simulation
- Data
- Graph

Simulation Graphs

traci5 vs traci6 Comparison

Select Map & Run

Map: Map2

Run: run_2026-04-13_151555_q_le

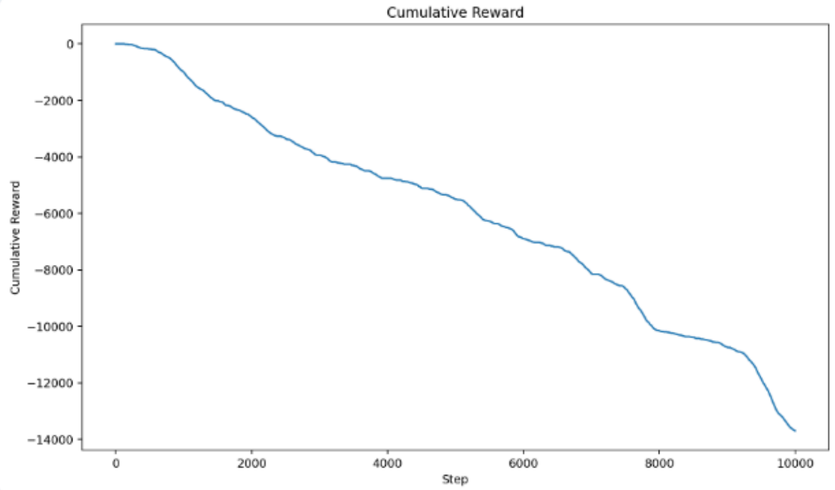
Open Plots Folder

Tip: Select a map, then a run folder (run_..._RL), then plots will appear.

Plots

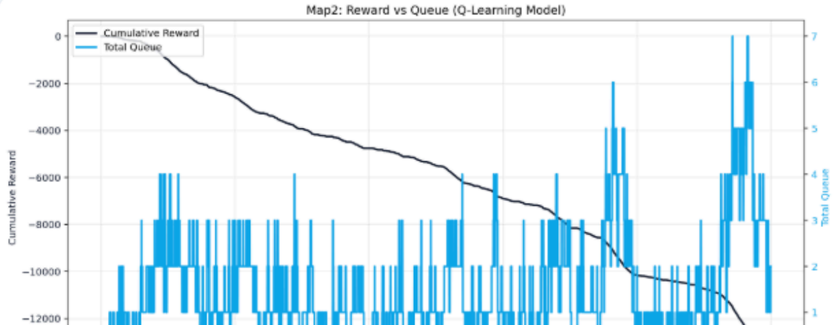
cumulative_reward.png

Open



dual_axis_reward_vs_queue.png

Open



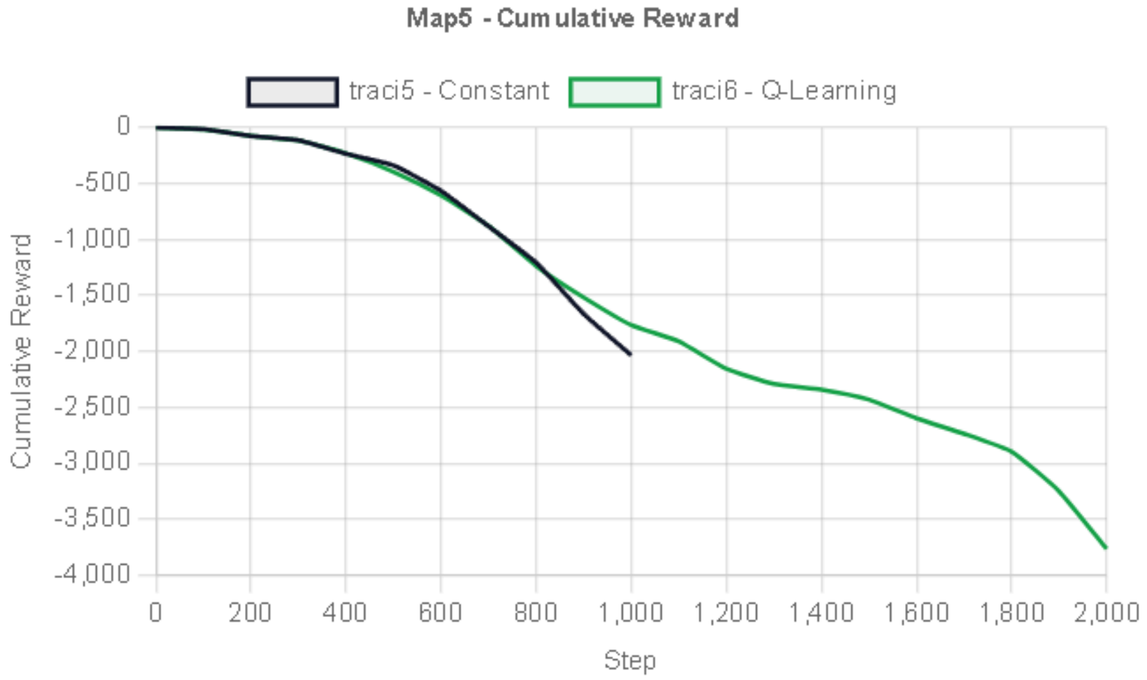
Map5

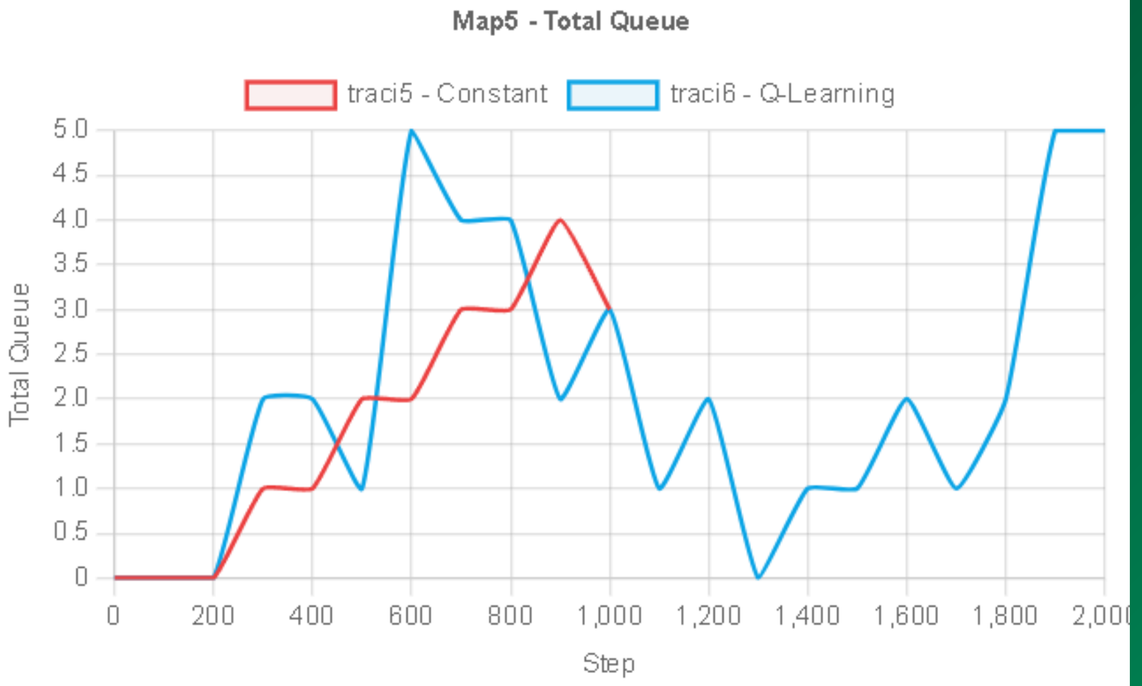
traci5 Run | traci6 Run

Winner: **Constant** | Queue Gain: **-0.32** | Reward Gain: **-1731.00**

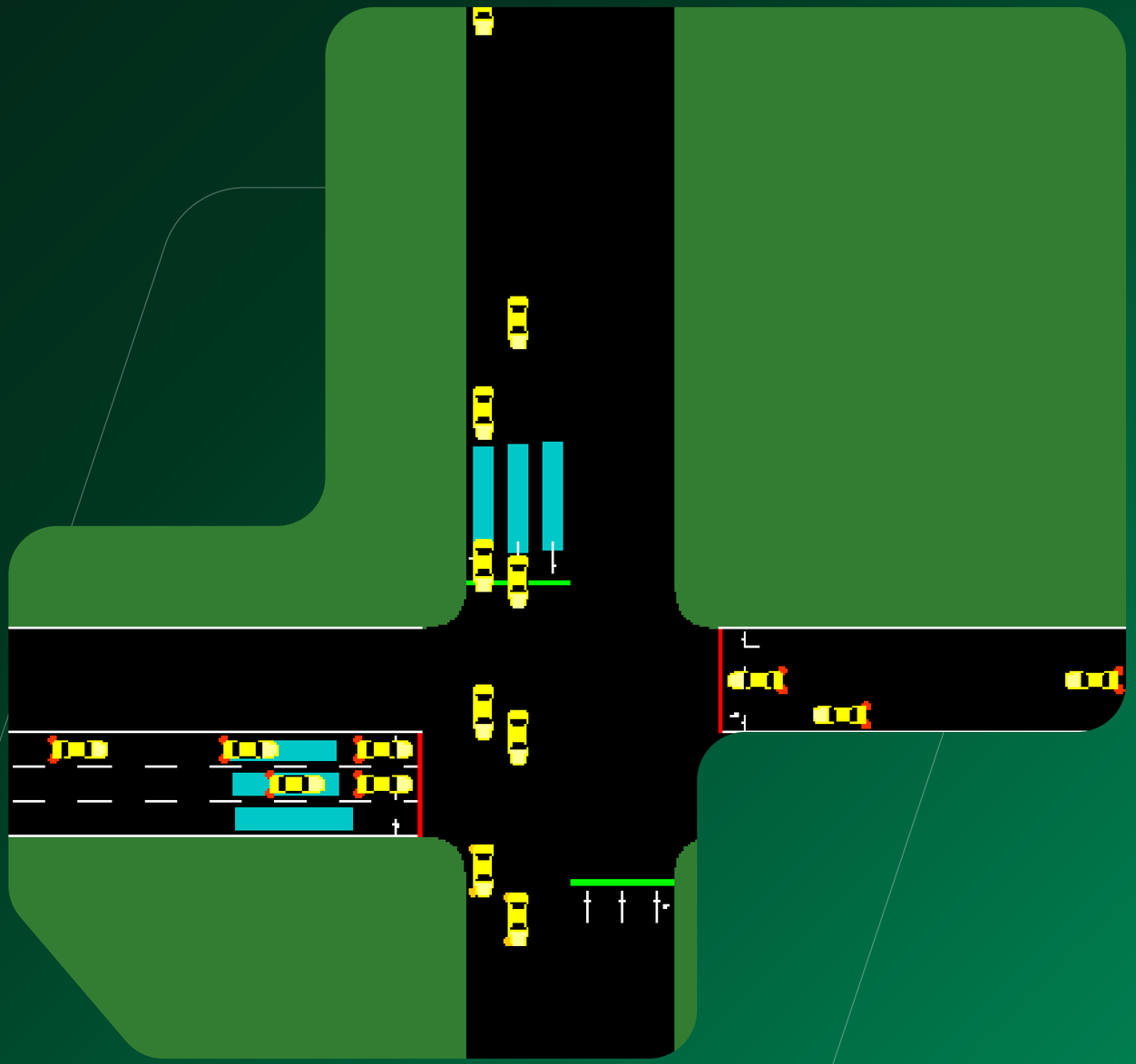
Constant Model: FAILED | Q-Learning Model: STOPPED

traci5 Avg Queue	traci6 Avg Queue	traci5 Final Reward	traci6 Final Reward
1.73	**2.05**	**-2036.00**	**-3767.00**

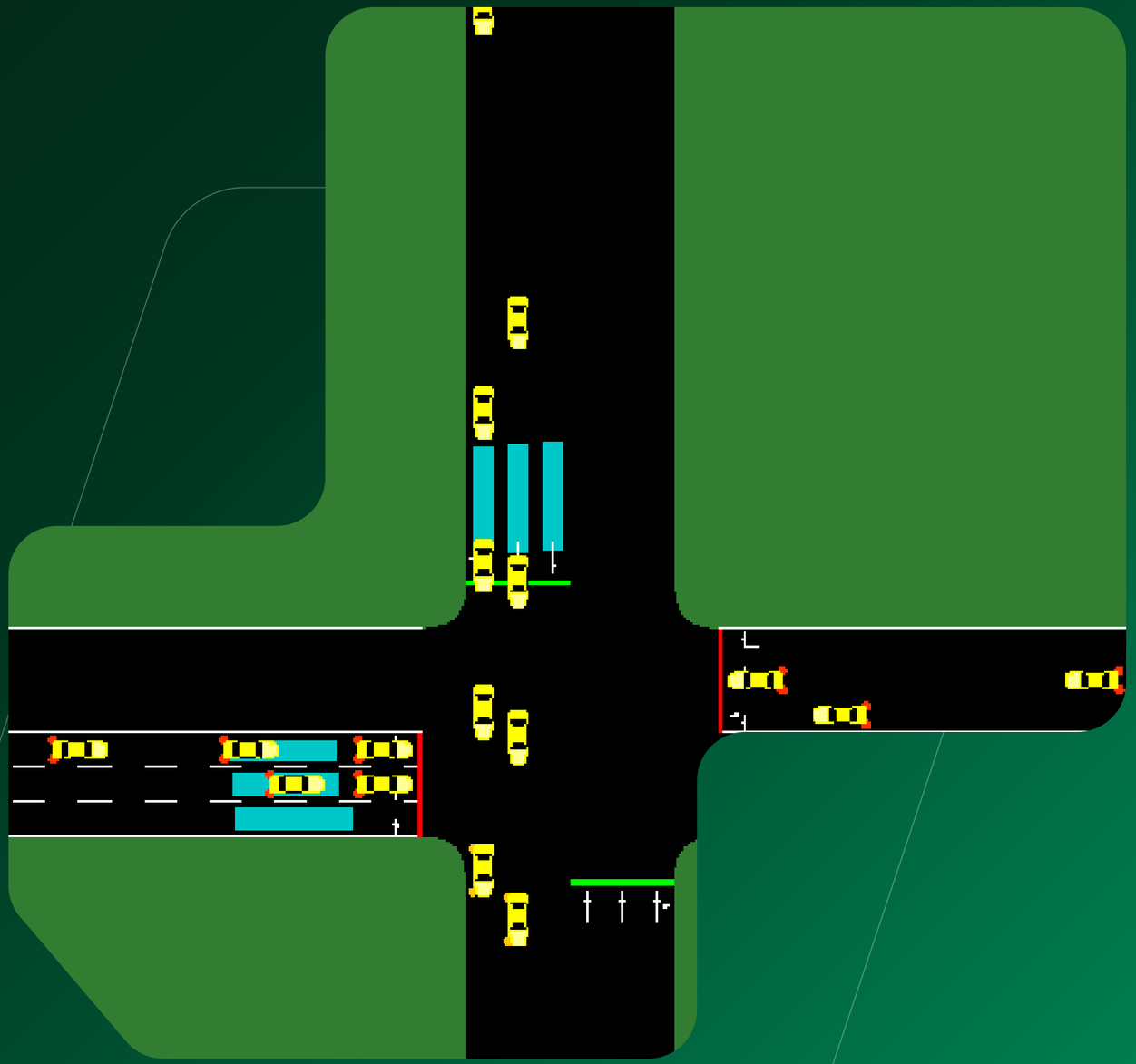




LIVE DEMO



LIVE DEMO



THANK YOU

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